Do School Size and Staffing Add to Socioeconomic Predictors of School ACT Scores?

Abstract

Socioeconomic factors explain most of the variations in high school average ACT scores and school size and staffing add a small but statistically detectable increment.I combined EdGap census-tract socioeconomic data with National Center for Education Statistics (NCES) enrollment and teacher counts in 20 US states (2016-17). Ordinary least squares models of average ACT used five socioeconomic predictors, then added log enrollment and student-teacher ratio. The added predictors raised R^2 from 0.628 to 0.632 and lowered mean absolute error from 1.145 to 1.137 ACT points.Free and reduced-price lunch share was by far the strongest predictor (standardized coefficient −1.78 ACT points per standard deviation).larger schools and schools with more students per teacher had slightly higher ACT scores, and both associations persisted after controlling for state and charter status. Once community context is known, school size and staffing add little predictive value.

Introduction

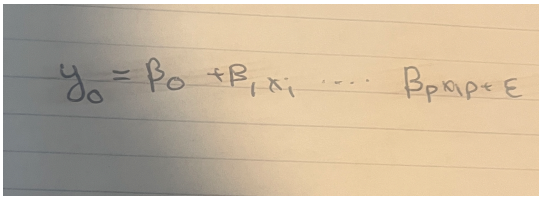
Student achievement is strongly tied to family and neighborhood socioeconomic status. Meta-analyses and national studies consistently find that students from poorer families and communities score lower on standardized tests (Coleman et al., 1966; Sirin, 2005; Reardon, 2011).A practical question follows: after accounting for community context, do features that districts can change, such as how large a school is and how many students each teacher serves, still relate to test performance? The education production-function literature is divided on whether staffing and resource measures matter (Hanushek, 2003; Krueger, 1999), and evidence on school size is mixed (Leithwood & Jantzi, 2009).

This project uses two public data sources. EdGap.org supplies 7,986 US schools with average ACT score (or SAT equivalent), the share of students eligible for free or reduced-price lunch, and four census-tract measures: unemployment rate, share of adults with a college degree, share of children in married-couple families, and median household income. The NCES Common Core of Data (CCD) for 2016-17 supplies each school's state, level and charter status (school directory), total enrollment (membership file) and full-time-equivalent teacher count (staff file). We joined the files on the NCES school identifier, keeping every EdGap school (left joins), and kept high schools only (7,230). Out-of-range values were set to missing, the 3 schools with no valid ACT score were dropped, and the remaining gaps in predictors (at most 24 values in any column) were imputed. The final data set contains 7,227 high schools in 20 states.

The aim of this project is to determine (1) how well socioeconomic factors predict a school's average ACT score and (2) whether school size, measured by log enrollment, and staffing, measured by students per teacher, improve that prediction beyond the socioeconomic factors.

Theoretical background

Linear regression. Ordinary least squares (OLS) models a school's average ACT score as a linear combination of predictors plus an error term, and chooses the coefficients that minimize the sum of squared residuals:



Each coefficient is the expected change in ACT for a one-unit increase in that predictor, holding the others fixed. Confidence intervals and p-values test whether a coefficient differs from zero. These tests assume independent errors with constant variance.

Standardized coefficients. Predictors here have very different scales (income in dollars, others as proportions). Converting each predictor to a z-score (subtract its mean, divide by its standard deviation) makes coefficients comparable: each then measures the ACT change for a one-standard-deviation increase.

Transformations. Enrollment is strongly right-skewed. Its natural logarithm reduces the influence of very large schools and expresses effects in proportional terms, so a doubling of enrollment has the same effect at any school size.

Missing data. Dropping every school with any missing predictor discards useful information. Multivariate imputation by chained equations predicts each incomplete variable from the others in repeated rounds (van Buuren & Groothuis-Oudshoorn, 2011). We used the iterative imputer in scikit-learn (Pedregosa et al., 2011), and left the outcome out of the imputation so that ACT could not influence the imputed predictors.

Methodology

All steps were run in a single Python notebook (pandas, scikit-learn, statsmodels; Seabold & Perktold, 2010).

Data preparation

1. Loaded the EdGap file, the 2016-17 CCD school directory, the CCD membership file (long format: one row per grade, race and sex) and the CCD staff file (one row per school).
2. Kept year, school ID, state, ZIP code, school type, level and charter status from the directory. Renamed all columns to lowercase snake_case.
3. Reduced membership to one row per school by keeping the "Education Unit Total minus Adult Education Count" row, and took teacher full-time equivalents (`TEACHERS`) from the staff file.
4. Cast the school ID to a common integer-like type in every file and left-joined directory, enrollment and teacher counts onto EdGap by school ID (7,986 rows retained).
5. Set impossible values to missing: ACT below 1 or above 36, free/reduced lunch share below 0 or above 1, and enrollment or teacher counts of zero or less.
6. Kept high schools only (7,230 schools; no duplicate rows or school IDs were found).
7. Created derived variables: student-teacher ratio (enrollment divided by teacher FTE, with ratios above 60 or below 3 set to missing), log enrollment, and a large/small school indicator (above or below median enrollment) used for plotting.
8. Dropped the 3 schools with no ACT score. Imputed the remaining missing predictor values with scikit-learn's `IterativeImputer` (random_state = 0), fitted on the seven numeric predictors without ACT. Clipped imputed proportions to [0, 1] and the ratio to at least 3.
9. Exported the cleaned data to `education_clean.csv` (7,227 rows).

Analysis

1. Examined distributions, a correlation matrix and scatterplots of ACT against log enrollment and student-teacher ratio.
2. Fitted a baseline OLS model of average ACT on unemployment rate, college share, married-couple share, median income and free/reduced lunch share.
3. Fitted a full model adding log enrollment and student-teacher ratio.
4. Compared the models with R^2, adjusted R^2, AIC, mean absolute error (MAE), root mean squared error (RMSE) and a nested F-test.
5. Refitted the full model on z-scored predictors to compare effect sizes, with 95% confidence intervals.
6. Inspected residuals against fitted values.
7. Checked robustness by adding state and charter status as categorical controls to the full model.

Computational Results

	enrollment	teacher_fte	student_teacher_ratio	log_enrollment
count	7227.00	7216.00	7206.00	7227.00
mean	1020.84	62.33	16.18	6.65
std	759.23	43.25	3.90	0.78
min	48.00	0.50	4.25	3.87
25%	422.00	28.21	13.56	6.05
50%	801.00	50.82	15.88	6.69
75%	1443.00	86.51	18.42	7.27
max	5000.00	291.00	57.20	8.52

Fig 1:Summary of school size and staffing variables

model	R-squared	Adj. R-squared	AIC	MAE	RMSE
socioeconomic only	0.6280	0.6277	26666.9917	1.1453	1.5299
socioeconomic + size + staffing	0.6319	0.6316	26593.8501	1.1365	1.5217

Fig 2:Fit of the socioeconomic-only and full models

```
Covariance Type:            nonrobust
================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept         21.4723      0.201    106.931      0.000      21.079      21.866
rate_unemployment -2.4866      0.404     -6.151      0.000      -3.279      -1.694
percent_college    1.4597      0.160      9.123      0.000       1.146       1.773
percent_married    0.0069      0.134      0.052      0.959      -0.256       0.269
median_income  -8.562e-07   1.23e-06     -0.697      0.486   -3.26e-06    1.55e-06
percent_lunch      -7.6042      0.096    -78.903      0.000      -7.793      -7.415
log_enrollment      0.1667      0.026      6.408      0.000       0.116       0.218
student_teacher_ratio  0.0176      0.005      3.516      0.000       0.008       0.027
================================================================================
```

Fig 3:Full model coefficients

	coef	ci_low	ci_high	p_value
rate_unemployment	−0.1407	−0.1855	−0.0958	0.0000
percent_college	0.2407	0.1890	0.2925	0.0000
percent_married	0.0013	−0.0489	0.0515	0.9587
median_income	−0.0209	−0.0795	0.0378	0.4858
percent_lunch	−1.7811	−1.8254	−1.7369	0.0000
log_enrollment	0.1293	0.0897	0.1688	0.0000
student_teacher_ratio	0.0683	0.0302	0.1063	0.0004

Fig 4: Standardized coefficients

```
                        coef  p_value
log_enrollment         0.1375      0.0
student_teacher_ratio  0.0383      0.0
R-squared with state + charter: 0.684
```

Fig 5: Full model with state and charter status added as controls

Discussion

Socioeconomic factors alone account for most of the variation in school ACT scores, and school size and staffing add only a little. The baseline model explains the 62.8% of the variance and the two added predictors raise this to 63.2 percentage points.Statistically there is an improvement.Practically it is small, since mean absolute error improves by approx only 0.009 points (Fig 2) .With more than 7000 schools even a small effect is easy to detect so statistical significance should not be read as practical importance.

Free and reduced-price lunch eligibility has an overwhelming association with the outcome. Holding all other covariates constant, a one-standard-deviation increase in the percentage of students qualifying for free or reduced lunch is associated with a 1.78-point decrease in the average ACT score. This standardized effect is roughly seven times larger than the next-largest predictor in the model, making socioeconomic status the dominant factor in explaining performance variation.

Income and married-couple share do not provide statistically significant unique information to the model once other factors are controlled for. Their intervals include zero ($p = 0.4858$ and 0.9587).This does not mean income is unrelated to ACT. Census-tract income, adult education and lunch eligibility describe overlapping aspects of community advantage, so after lunch share and college share are in the model, income and family structure carry little additional information.

School size and staffing have small positive associations. Doubling enrollment is associated with about 0.12 more ACT points ($0.167 \times \ln 2$), and each additional student per teacher with about 0.02 more points. The staffing sign is the opposite of what a simple class-size argument predicts, and it should not be read as evidence that crowding helps. Larger schools may offer more advanced courses, and staffing ratios may differ by school type and location in ways the model does not capture. Adding state and charter status raised R^2 to 0.684 and roughly doubled the staffing coefficient (0.018 to 0.038), while the enrollment coefficient was similar (0.167 to 0.138), so both associations survive but their size depends on which comparisons are made.

Some limitation are

- The data cover only 20 states and states differ by how many students take the ACT so average scores are not comparable in selectivity.
- Models were evaluated on the same schools used to fit them, so the improvement from added predictors may be slightly optimistic. A held-out test would be the natural next step.
- The results are associations from one school year (2016-17), not causal effects. Lunch eligibility is an imperfect poverty measure (Harwell & LeBeau, 2010).
- A few percent of predictor values were imputed, and the student-teacher ratio depends on reported teacher full-time equivalents and on cutoffs (3 and 60) that I chose.

Conclusion

This study asked whether what a school looks like on the inside, its size and teacher staffing, helps explain test scores once the surrounding community is accounted for. Across 7,227 high schools, community socioeconomic conditions predicted about 63% of the variation in average ACT scores, led by the share of students receiving free or reduced-price lunch. Adding school size and student-teacher ratio improved the fit by a statistically detectable but practically small amount (R^2 from 0.628 to 0.632, error reduced by 0.009 ACT points), and both predictors remained positively associated with ACT after controlling for state and charter status.

For policy, the results suggest that differences in ACT scores between schools mostly mirror differences between their communities, and that changing school size or staffing alone is unlikely to close those gaps. Because the findings are associations from 20 states and one school year, they are best treated as a description of how scores vary, not as a guide to what any single school would gain from changing its size or ratio.Future work could test the models on held-out schools, account for clustering within states and districts, and add school resources that may matter more, such as per-pupil spending or course offerings.

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